3019061 ITNPAI3 Assignment 2 Report

This assignment investigated sentiment analysis using two neural approaches on the Kaggle Sentiment Analysis Dataset with three classes: negative, neutral and positive. The aim was to move beyond the Bag-of-Words and TF-IDF features used in Assignment 1 and compare two neural input representations and two different neural architectures. The selected approaches were: (1) a BiLSTM with trainable word embeddings learned from scratch, and (2) a DistilBERT classifier using pretrained contextual embeddings. The comparison focuses on preprocessing for neural models, model design, performance on a held-out test set, and analysis of common misclassification patterns.

Task 1: Preprocessing for neural sentiment models

Preprocessing was deliberately light and sentiment-aware. In sentiment analysis, aggressive cleaning can remove useful cues, especially in social media text. Emojis and emoticons were preserved by converting them into descriptive tokens rather than deleting them. This was important because symbols such as ":)", ":(" or emojis often carry sentiment directly and can strengthen polarity.

Negation and contractions were also preserved. For the word-level BiLSTM pipeline, contractions such as "don't" and "isn't" were expanded to "do not" and "is not" so that the negation term remained explicit. This helps the model learn that negation can reverse sentiment. Earlier feedback suggested that some stopwords could be removed to reduce vocabulary size, but in this assignment neutral stopword removal was disabled because neural models benefit from preserving syntax and word order. This is particularly important for negation, contrast and emphasis.

Two tokenisation strategies were used because the two models require different input formats. The BiLSTM used word-level tokenisation, which is appropriate for a trainable embedding layer. DistilBERT used subword tokenisation through the pretrained DistilBERT tokenizer. This is preferable for transformer models because subword units handle slang, misspellings and rare words more effectively.

Sequence padding and truncation were based on the 95th percentile of the training length distribution, rather than using an arbitrary fixed length. This limited excessive padding while preserving most sentences. Out-of-vocabulary words were handled differently in the two pipelines. The BiLSTM used an <OOV> token, while DistilBERT handled unknown words naturally through subword splitting.

Finally, case normalisation and punctuation were treated carefully. Lowercasing was applied to reduce vocabulary sparsity, which was appropriate because the model used distilbert-base-uncased and a lowercased word-level pipeline. Useful punctuation such as "!" and "?" was retained because it often indicates intensity or emphasis. URLs and usernames were normalised to placeholder tokens, and hashtag content was preserved rather than discarded.

Task 2: Neural input representations

The first representation was a trainable embedding layer learned from scratch in the BiLSTM model. This approach allows the model to learn domain-specific sentiment cues from the training data. Its advantage is simplicity and efficiency, but it is limited because each word receives only one static vector regardless of context.

The second representation was DistilBERT's pretrained contextual embeddings. These embeddings are context-sensitive, meaning that a word's representation depends on the sentence in which it appears. This is especially valuable for sentiment analysis because polarity often depends on negation, contrast, sarcasm, or domain-specific usage.

Overall, the contextual representation captured sentiment better than the trainable embedding. This is reflected in the test results: DistilBERT achieved higher overall accuracy and F1, and it outperformed the BiLSTM in every class. The difference suggests that pretrained contextual knowledge is particularly helpful for short informal text, where meaning is often subtle and context-dependent.

Task 3: Neural architectures and results

The first model was a BiLSTM with trainable embeddings. The main hyperparameters were: embedding dimension 128, LSTM units 64, dense layer size 64, dropout 0.3, batch size 32, learning rate 0.001, and up to 8 epochs. Early stopping with patience 2 was used to reduce overfitting.

The second model was DistilBERT for sequence classification. The main hyperparameters were: batch size 16, learning rate 2e-5, and 2 fine-tuning epochs. The best checkpoint was selected using validation F1. This model also benefited from the regularisation built into the pretrained transformer architecture.

The dataset was split using train.csv for training and validation, and test.csv as the final held-out test set. Accuracy and F1 were used for evaluation. Because the task is three-class sentiment classification, macro F1 is especially useful as it gives equal importance to each class.

Model	Representation	Architecture	Accuracy	Macro F1
LinearSVC	TF-IDF / BoW	Linear SVM	0.7006	0.7032
BiLSTM	Trainable embeddings	BiLSTM	0.710	0.712
DistilBERT	Pretrained contextual embeddings	Transformer	0.787	0.792

The best traditional model from Assignment 1 was TF-IDF with LinearSVC, which achieved an accuracy of 0.7006 and a macro-F1 of 0.7032. The BiLSTM with trainable embeddings slightly improved on this baseline, reaching 0.7100 accuracy and 0.7120 macro-F1. However, the strongest performance came from DistilBERT, which achieved 0.7870 accuracy and 0.7920 macro-F1. DistilBERT improved accuracy by around 7.7 percentage points and macro F1 by around 8.0 percentage points over the BiLSTM. This suggests that contextual embeddings

and transformer attention mechanisms capture sentiment more effectively than a sequence model trained from scratch.

Per-class performance supports the same conclusion. For the negative class, BiLSTM achieved an F1 of 0.68, whereas DistilBERT reached 0.80. For neutral, the scores were 0.69 and 0.76 respectively. For positive, the scores were 0.76 and 0.82. The largest gain was in identifying negative sentiment, indicating that DistilBERT handled more nuanced polarity shifts better than the BiLSTM.

A further point to note is the trade-off between performance and computational cost. Although DistilBERT achieved the strongest results, it required substantially more training time and memory than the BiLSTM and the traditional models from Assignment 1. This means that model choice should depend not only on accuracy, but also on practical constraints such as hardware, training time, and deployment requirements. Another limitation is that sentiment labels can be inherently subjective, especially for neutral or mixed-sentiment examples, which may reduce the upper bound of achievable performance. Future work could explore stronger hyperparameter tuning, class-balancing strategies, pretrained static embeddings such as GloVe, and data augmentation for rare or ambiguous sentiment patterns. More advanced error analysis could also examine whether certain classes or linguistic constructions are consistently harder across models.

Task 4: Misclassification analysis

Misclassification analysis showed that the neutral class was the hardest to separate cleanly from the other two classes. The BiLSTM frequently confused negative with neutral and positive with neutral, while DistilBERT reduced these errors but did not eliminate them. This suggests that many difficult cases are not strongly polarised.

Five recurring error patterns were identified. First, negation remained challenging. Phrases such as “not good” or “wasn’t bad” can reverse or soften polarity, and the model must learn the scope of the negation rather than rely on isolated sentiment words. Second, mixed-sentiment sentences were difficult because they contain both positive and negative content, often linked by words such as “but” or “however”. In such cases, overall polarity may be ambiguous even for humans.

Third, sarcasm and irony were problematic. These expressions often use positive surface words to communicate negative meaning, which is difficult without pragmatic or world knowledge. Fourth, long-range dependencies created errors when the true sentiment depended on information appearing later in the sentence. Although BiLSTM and DistilBERT are both designed to handle sequences, delayed polarity shifts still remain challenging. Fifth, domain-specific usage caused mistakes because some words change sentiment depending on context. For example, words such as “sick”, “brutal” or “unreal” may be positive in one domain and negative in another.

These errors are difficult for neural models because sentiment is not determined only by individual words. It often emerges from composition, context and implied meaning. DistilBERT performed better on these cases because contextual embeddings and self-

attention provide richer sentence-level representations, but even this model still struggled with sarcasm, weak sentiment and class overlap.

Conclusion

This assignment showed that neural models provide a clear improvement over simpler text representations for sentiment analysis. The BiLSTM with trainable embeddings was a reasonable neural baseline and performed moderately well, but DistilBERT was clearly stronger. The contextual embedding approach captured sentiment more effectively, especially for negative and ambiguous cases. Preprocessing was a major factor in this success: preserving emojis, negation, punctuation and informal textual cues helped both models, while model-specific tokenisation ensured the inputs matched each architecture. Overall, DistilBERT was the best model because it combined pretrained linguistic knowledge with contextual understanding, although challenging cases such as sarcasm, mixed sentiment and domain-specific language remain open problems.